Dynamic Systems Thinking for Psychologists

Assessment 2026: Individual Report and Oral Exam

At a glance

- **What:** an individual written report in which you describe, formalise, simulate and critically discuss a dynamic system of two or more psychological constructs. It starts from the part of the system you **built with your group** and goes beyond it, either by **extending the model** or by **designing and testing it** (Section 2).
- **Length:** 1000–3000 words of main text. Title page, references, figure captions, equations, tables and the code appendix do **not** count. With figures this is typically 7–15 pages.
- **Submit:** by email to michael.aristodemou@psychologie.uzh.ch by **11 December 2026, 23:59**. Attach (1) the report as **PDF** and (2) your **R code** as one `.R` or `.Rmd` file that runs from top to bottom. Name the files `Lastname_Firstname_DSTP_report.pdf` and `Lastname_Firstname_DSTP_code.R`.
- **Oral exam:** a 15–20 minute conversation about *your* report (Section 6).
- **Grade:** report and oral exam together make up 100% of the grade (1–6 scale, half grades).

1 The task

Choose a set of **two or more psychological constructs that change over time and influence each other**. Together they form the system you study. Tell a story about how these constructs change over time, how they affect each other, and how people might differ in these changes. Then make this story precise: draw it, write it as equations, simulate it in R, and discuss what the model can and cannot tell us.

Your story does not have to be supported by empirical research. It has to be a *plausible and well-argued* description of one way the system could work. Do not let your technical skills limit your thinking: you may describe parts of the system that you cannot formalise or simulate. Then say what is missing and why you chose the simplified version.

"Influence each other" means that the constructs are **coupled**: the state of one construct appears in the equation for the change of another (for example, worry lowers sleep quality, and poor sleep increases worry). A single process that only reacts to an outside input (for example, tiredness that only responds to the hours slept) is not enough on its own. You can include such inputs, but at least two constructs need their own equations and at least one coupling between them.

The report should show that you understand the course concepts and can think about a psychological phenomenon from a dynamic systems perspective.

2 Start from your group's model, then go further

During the course, you build part of your system together with your group. Your report **starts from this group model**. Summarise it briefly (about 300–500 words); do not re-explain all of the group code line by line. You may reuse the group's code and figures. Then go beyond the group model individually, in one of two ways. Say which route you take at the end of your introduction.

Whichever route you choose: one part that is worked out carefully (motivated, formalised, simulated, interpreted, criticised) is better than several shallow ones.

2.1 Route A: Extend the model

Change or add to the theoretical model, and show what the change does to the behaviour of the system. Examples:

1. **A new construct or feedback loop:** add a third variable, or a coupling in the opposite direction.
2. **Individual differences:** let people differ in their parameters (e.g., resting level B, return speed A, coupling strength A_{cross}). Explain what each difference means psychologically and show how trajectories differ between people.
3. **Interventions or events:** add a shock or input at a specific time (e.g., a therapy session or a life event). Compare scenarios: one-off vs. repeated, early vs. late, small vs. large.
4. **Non-linearity:** thresholds, floors and ceilings, or a coupling that changes with the state of the system. Show what the linear version would miss.

You can **optionally** add any of the following to a Route A report:

5. **Measurement:** simulate how you would observe the system (measurement noise, measurement interval) and show how conclusions depend on it (Section 4.4).
6. **Designing an experiment:** describe a study that could test your extended model (a shorter version of Route B, Section 4.4).
7. **Fitting:** fit a model (e.g., with `ctsem`) to data you simulated and check whether you recover the true parameters (Section 4.5).

2.2 Route B: Design and test the model

Keep the group model as it is (or change it only slightly), and work out in detail **how it could be tested empirically**. Because the theoretical model does not change, this part must be **substantial**. It should include:

- **A complete study design:** who takes part, what you measure with which instruments, how often and over what period, and, if you manipulate something (e.g., an intervention), what and when.
- **A link to the dynamics:** why this measurement interval and study length, given how fast your constructs change and how long their effects last?
- **Simulated data from the design:** generate the data your study would produce from the model, including measurement error.

- **An analysis of these data**, done as you would analyse real data (e.g., a cross-lagged regression or a `ctsem` model). Show whether the design can detect the effects you care about, for example by comparing estimates with the true values, or by comparing designs (different intervals, numbers of measurements or participants).
- **Interpretation:** which results would support the model, and which would speak against it?

3 Recommended structure

You may organise the report differently, but it must cover all of the points below. The word counts are rough guides.

Section	What it should contain	Words
1. Introduction	The phenomenon, why it is interesting as a *dynamic* system, your **research question** (what should the model explain or predict?), and which route you take.	300–500
2. Verbal model	The story in words, with a **path diagram** (boxes for constructs, arrows for effects, sign on every arrow). Which loops reinforce, which dampen? What is left out, and why?	300–500
3. Formal model	The equations, with every symbol defined, plus a **parameter table** (see Section 4.1).	200–500
4. Simulation results	Figures that answer your research question: baseline behaviour, scenarios, a parameter you varied, individual differences. Every figure is interpreted in the text.	400–800
5. Study design and measurement	Route A: optional (up to about 300 words). Route B: the main part, about 600–1000 words (see Section 2.2).	see left
6. Discussion	What did you learn? Specific limitations, what you could not model or test, and what would come next.	300–600
References	APA style.	–
Appendix	Full R code (as text, not screenshots).	–

4 What we expect in each part

4.1 The formal model

- **Write the model as change over time.** Use the notation from the course, e.g.

$$\frac{d\eta_1}{dt} = A_1\,\eta_1(t) + A_{cross_{12}}\,\eta_2(t) + B_1 \quad (+ \text{ noise } G_1\,dW_1)$$

  Everything on the right is evaluated at the **same time** t. Do not mix in discrete-time notation such as η_{t-1}.
- **Say what time is.** Give the unit (hours, days, weeks) and the time span you simulate. The time axis must be real time, not a dose level or a trial number.
- **Define every symbol** and say what it means psychologically.

- **State the direction of each construct:** what does a higher value mean (more worry, better sleep)? The numbers themselves can be on an arbitrary scale; the direction is what makes the signs of your couplings interpretable.
- **Give a parameter table** with the symbol, its meaning, the value you used, and its **direction and relative strength** (e.g., "worry harms sleep more than sleep reduces worry"). You do not need to justify the precise numbers.
- **Check your model.** Without noise, does the system come to rest after a disturbance, or does it run away? Is this what your story predicts?

Example (worry η_1 and sleep quality η_2, arbitrary units, higher = more worry / better sleep, time in days):

$$\frac{d\eta_1}{dt} = A_1\,\eta_1 + A_{cross_{12}}\,\eta_2 + B_1, \qquad \frac{d\eta_2}{dt} = A_{cross_{21}}\,\eta_1 + A_2\,\eta_2 + B_2$$

Symbol	Meaning	Value	Direction and relative strength
A_1	How fast worry returns to its resting level	-0.5	Negative: worry returns to rest. Slower than sleep quality.
A_2	How fast sleep quality returns to its resting level	-1.0	Negative: sleep quality returns to rest. Faster than worry.
$A_{cross_{12}}$	Effect of sleep quality on worry	-0.3	Better sleep reduces worry. The weaker of the two couplings.
$A_{cross_{21}}$	Effect of worry on sleep quality	-0.4	More worry makes sleep worse. Together with $A_{cross_{12}}$ this forms a reinforcing loop.
B_1, B_2	Intercepts	3.3, 7.2	Set the resting levels (here worry = 3, sleep quality = 6).

4.2 Figures

- Every figure has a **number, a caption, labelled axes** (with the time unit), and a legend where needed.
- Every figure is **interpreted in the text**: what do we see, and why does the model produce it? Refer to the equations ("because $A_{cross_{21}}$ is negative, ...").
- Start simple: show the system **without noise** first, so that the deterministic behaviour is visible, then add noise.
- Show **more than one person** when you talk about individual differences, and **more than one scenario** when you make a comparison.

4.3 Simulations

- Show at least: (a) the baseline behaviour of the system, (b) the scenario(s) that answer your research question, and (c) **what happens when you change at least one key parameter** (a "what if" analysis), with an explanation.

- If you **build a pattern directly into the equations** (e.g., an oscillation or a jump at a fixed time), say so clearly. The pattern then comes from your input, not from the dynamics of the system. Discuss this.

4.4 Study design and measurement

Optional for Route A; the main part of Route B (see Section 2.2 for what it should include).

Link your model to data. Which instrument would measure each construct? How often would you measure, and why does the measurement interval matter for your system (think of how fast your constructs change)? Which part of the variation is measurement error, and which is real change (process noise)? If you design an experiment, say what you manipulate, when, and what pattern of results you expect.

4.5 Fitting (optional)

If you fit a model to simulated data, report the **true and estimated values side by side** and discuss differences. If you cannot recover the parameters, try to explain why (e.g., the fitted model differs from the one that generated the data, too few time points, the measurement interval is too long for your process). A careful explanation of a failed recovery is worth as much as a successful one.

4.6 Discussion

Avoid generic limitations ("the model is a simplification", "more variables could be added"). Name **specific** ones and their consequences: *"The coupling is assumed to be the same at all levels of worry. If worry only disturbs sleep above a threshold, the model would overestimate . . . ".*

4.7 R code

- The code runs **from top to bottom** in a clean R session and reproduces every figure in the report (use `set.seed()`).
- All parameters are defined **at the top**, with a comment explaining each one.
- Comment what each block does and why, not only what the R function does.
- In the main text, show only **short, key snippets** (e.g., the lines that compute the slopes). The full code goes in the appendix and in the separate code file. Do not paste screenshots of code.

```
# --- Parameters (time unit: days) ----------------------------------------
A1  <- -0.5   # return speed of worry (slower than sleep quality)
A12 <- -0.3   # effect of sleep quality on worry (better sleep, less worry)
...
# --- Slopes, computed from the CURRENT states in every small step -------
dWorry <- A1 * worryState + A12 * sleepState + B1
dSleep <- A21 * worryState + A2 * sleepState + B2
```

5 Assessment criteria

The report and the oral exam are graded on the following criteria. The right column describes what a strong report does.

Criterion	A strong report …
1. Understanding of course content	uses course concepts (rate of change, return speed, resting level, coupling, process vs. measurement noise, individual differences, measurement interval) correctly, and links each to the psychological story.
2. Complexity & depth	goes clearly beyond the group model, either by extending the model (e.g., multiple processes, time scales, individual differences) or by a detailed study design and test of the model; shows original thinking about the phenomenon.
3. Critical thinking	considers strengths, alternatives and limitations; names specific limitations and their consequences.
4. System specification & analysis	states the model clearly in words, diagram and equations; explains the direction and relative strength of the parameters; uses simulations and "what if" analyses that answer the research question; visualises the dynamics clearly.
5. Clarity & organisation	has a clear research question, logical flow, informative headings, numbered and captioned figures, and stays within the word limit.
6. Use of R code	provides code that runs, reproduces all figures, is commented and organised, with parameters defined in one place.

6 Oral exam

The oral exam is part of the same assessment component as the report. It is a 15–20 minute conversation, and all questions concern the content of **your** report. Be prepared to explain every equation, figure and line of code in your report.

7 Grading

The grading scale for the course is 1–6, with half grades. The individual report and the oral exam together make up 100% of the final grade. The assessment criteria above are the basis for grading both the report and the oral exam.

8 Frequently asked questions

Do I need real data? No. Simulated data are enough, also for Route B.

Do I have to change the group's model? No. If you keep it, take Route B: a detailed study design and a test of the model on data simulated from that design.

Can I use a non-linear model or a model we did not cover? Yes, if you explain it clearly. Also discuss why the linear version would not be enough.

What if I cannot implement part of my idea? Describe it in words or as an equation, explain what stops you from simulating it, and discuss how the simplified version you did simulate differs from it. This counts positively under critical thinking.

Can I reuse the group's code and figures? Yes. You do not need to mark which code comes from the group.

9 Submission checklist

- ☐ My introduction states my research question and which route (A or B) I take.
- ☐ My path diagram shows all constructs and a sign on every arrow.
- ☐ Every equation is written as change over time, and every symbol is defined.
- ☐ I state the time unit and what higher values of each construct mean.
- ☐ My parameter table gives the meaning, direction and relative strength of each parameter.
- ☐ Every figure has a number, a caption and axis labels, and is discussed in the text.
- ☐ I show individual differences and at least one "what if" parameter change.
- ☐ Route B only: I include a complete study design, simulated data from it, and an analysis of these data.
- ☐ My limitations are specific to my model.
- ☐ The main text is 1000–3000 words.
- ☐ My code runs from top to bottom and reproduces all figures.
- ☐ I send the PDF and the code file by 11 December 2026.